

Kinora TMJ

Clinical AI Orientation (Part 14)

Mandible, Disc, Mastication Muscles, Mechanics & Disorders

Version 1.0 — Kinora Admin Conocimientos Upload

Disclaimer

Educational only. Screen odontogenic infection, temporal arteritis red flags, and neurologic deficits. TMJ care does not replace dental/medical assessment.

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1. Bones & Disc

RECORD: Mandible

ID: TMJ-BONE-01

Landmarks: Condyle, coronoid, ramus, body, angle, lingula

Articulations: Temporal bone via disc (TMJ)

Muscles: Masseter, temporalis, pterygoids, digastric, floor of mouth

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Temporal Bone (Mandibular Fossa)

ID: TMJ-BONE-02

Landmarks: Mandibular fossa, articular eminence

Clinical: Eminence guides translation; steep eminence → locking risk

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Articular Disc

ID: TMJ-BONE-03

Structure: Biconcave fibrocartilage; posterior bilaminar zone vascular/neural

Function: Congruence; shock; translation with condyle

Pathologies: Disc displacement with/without reduction; perforation

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

2. Ligaments

RECORD: TMJ Ligaments

ID: TMJ-LIG-01

Include: Temporomandibular (lateral) ligament, sphenomandibular, stylomandibular

Function: Limit excessive movement; lateral ligament resists posterior dislocation

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

3. Muscles

RECORD: Masseter

ID: TMJ-MUS-01

Origin: Zygomatic arch

Insertion: Ramus/angle

Innervation: CN V3

Function: Elevation (powerful); protrusion assist

Clinical: Bruxism hypertrophy; trigger points

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Temporalis

ID: TMJ-MUS-02

Origin: Temporal fossa

Insertion: Coronoid

Innervation: CN V3

Function: Elevation; posterior fibers retraction

Clinical: Temporal headache overlap

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Lateral Pterygoid

ID: TMJ-MUS-03

Origin: Sphenoid / lateral pterygoid plate

Insertion: Condyle neck + disc (superior head)

Innervation: CN V3

Function: Protrusion, contralateral excursion; disc control

Clinical: Key in disc displacement pathophysiology

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Medial Pterygoid

ID: TMJ-MUS-04

Origin: Pterygoid plate

Insertion: Medial ramus

Innervation: CN V3

Function: Elevation, protrusion, contralateral excursion

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

4. Biomechanics

RECORD: Opening Mechanics

ID: TMJ-BIO-01

Phases: Rotation in fossa early (~20–25 mm) then translation down eminence

Normal_ROM: Opening often ~40–50 mm interincisal (individual)

Clinical: Deviation, deflection, locking

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Closing / Chewing / Speech

ID: TMJ-BIO-02

Closing: Temporalis/masseter/medial pterygoid

Chewing: Alternate working/balancing sides

Speech: Fine mandibular control with hyoid/tongue

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

5. Pathologies

RECORD: TMD Myalgia / Arthralgia

ID: TMJ-PATH-01

Symptoms: Jaw pain, fatigue, headache, ear fullness mimic

DDx: Odontogenic, giant cell arteritis (age>50 temporal — urgent), otologic

Treatment: Education, soft diet brief, habit reversal, exercise, occlusal appliances selected evidence mixed

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Disc Displacement

ID: TMJ-PATH-02

With_Reduction: Click, intermittent catch

Without_Reduction: Closed lock, limited opening

Imaging: MRI gold for disc position

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Headache Related to TMD

ID: TMJ-PATH-03

Note: Overlap cervicogenic/migraine — screen SNNOOP10 red flags

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.